

The Carpet Industry Nudges Toward Green



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Carpets play a critical role in making homes more comfortable by warming up cold floors and reducing sound transfer between floors. Without carpets, unheated concrete slabs and strutting around barefoot in the winter would be intolerable.

Together, a carpet plus padding—an underlay or cushion—provides a small amount of insulating value to floors. This additional insulation equivalent is measured in “togs,” which is an assessment of heat flow resistance through cloth; an R-value of 3 is comparable to 1.7 togs. The Carpet and Rug Institute makes it a bit simpler by estimating a carpet’s R-value as the total carpet and underlay thickness measured in inches multiplied by 2.6. A combination of nylon carpet and a polyurethane cushion with a total thickness of 1.5 inches, for example, will yield an overall R-value of 3.9. This is roughly comparable to a ¾-inch rigid foam panel.

Still, carpets are not an entirely benign component of the home-building industry. The biggest issue is the disposal of used and worn-out carpet. In 2002, 2.25 million tons of old carpets were sent to the nation’s landfills; only 90,000 tons (3.8%) of discarded carpets were recycled or reused last year.

In response to the disposal issue, as well as to manufacturing concerns, the carpet industry has been pushing itself to become greener in recent years. Manufacturers of nylon filament that make up carpet yarn are making efforts not only to add recycled content to their fiber, but also to market products that would be acceptable to environmentally concerned consumers.

Fiber Profiles

Most of the carpet manufactured in the United States is made of one of five

common types of fiber: nylon, wool, acrylic, polyester, or polypropylene (olefin). At least 97% of the fiber used by the U.S. carpet industry is synthetic, with nylon representing about 65% of the carpet sold in the United States.

Nylon dominates the residential carpet industry for good reasons. It is durable; it is static-free; it maintains color; and it resists soiling, staining, and

Environmental Issues

Environmental concerns arise both in the manufacturing of carpet and with its disposal. One company that has started to address the environmental impact of carpet manufacturing is DuPont. It has developed a measure of sustainability for its production of Antron nylon fiber that



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mildew. Wool is considered luxurious and is quite strong, but it retains moisture, is not static resistant, can fray, and is expensive. Acrylic is an artificial fiber that looks and feels like wool but is much less expensive. Although it resists moisture, mildew, fading, crushing, staining, sun damage, and static electricity, acrylic fiber is not durable. Polyester is durable, easy to clean, resistant to some stains, and less expensive than wool and nylon. But it does fade with sunlight. Polypropylene is the next-best seller after nylon and is most commonly used for commercial carpet. Polypropylene fibers are colorfast, strong, mildew resistant, moisture resistant, and easy to clean—but can be easy to crush; it is the most inexpensive fiber.

it calls Total Environmental Impact. This scale measures progress in the means of manufacturing carpet relative to how it has been manufactured in the past. The scale ranges from 1 to 10, and includes the following factors: depletable natural resources; global warming emissions; other air emissions; land emissions (landfill and deep-well waste disposal); safety and health; and material recovery. It also includes water use. Water is assumed to be treated on-site, so this is water that is taken and not returned to its original source at least as clean and at the same temperature and pH as it was when it was originally obtained. Although self-rating is not always the most effective assessment technique, the basic structure of this program is based on life cycle

analysis, and the company has rated itself quite conservatively so far.

Other carpet makers are touting the environmental benefits of their final products. Some are beginning with recycled content; several means exist of incorporating recycled content into yarn and finished carpet.

Spinning Yarns

Synthetic filaments for carpet production are spun into fibers, which are then twisted together to form a continuous strand of yarn. Carpet fiber has two basic classifications: staple and bulked continuous filament (BCF). Staple fibers are made in short lengths; these are spun and twisted together to form long threads or yarn before being tufted into carpet. Often lower face weight products and higher-end carpet products are manufactured using staple yarns. BCF fibers are long filaments plied together to form continuous bundles of yarn. BCF fibers do not shed loose filaments following installation, as staple fibers tend to do. Most BCF ends up in commercial carpet, although as more recycled carpet is made, more BCF is ending up in residential carpets. Nylon is made in both BCF and staple fiber.

For the carpet industry, two types of filament are commonly produced—nylon 6,6 and nylon 6. Although the two appear similar, that similarity fades upon closer examination. Both are organic polymers, containing carbon, hydrogen, oxygen, and nitrogen atoms in monomer form; the molecules are linked to each other in a single-file chain. The numbering system that is used to distinguish different nylons refers to the number of carbon atoms between successive nitrogen atoms in the main monomer chain.

Depolymerization is the first step in recovering nylon as a recycled material. The recycled monomers are then combined with their virgin counterparts to create nylon that is spun into yarn with recycled content. Nylon 6 is made up of only one monomer (caprolactam), which

means that it can be chemically separated fairly easily, creating recycled monomers that can be reused. Nylon 6,6, on the other hand, is made up of two different monomers: hexamethylene diamine and adipic acid. Some carpet manufacturers prefer nylon 6,6 because it combines toughness with stiffness. However, its compound structure makes it more difficult to recycle. Although buyers who are interested in purchasing products with recycled content may be tempted to look



In Paris, old carpets can be recycled as flow-regulating devices in city sewers.

for nylon 6 carpets, neither type of nylon is inherently better than the other. In fact, manufacturers of nylon 6,6 have found ways of incorporating recycled content into the inorganic backing portion of their carpets.

Other manufacturers, such as BASF, make a bicomponent BCF fiber. Each filament of this fiber is structured with a nylon 6 core surrounded by a sheath of a nylon 6,12 homopolymer. In addition to the basic nylon polymer, which contributes up to 90% of the content, this fiber contains the homopolymer sheath, a lubricant, an antistatic additive, and a deluster additive. The nylon 6 polymers contain recycled caprolactam, which is reclaimed from a manufacturer's own production of nylon fiber.

Yarn can be made either as a colored product or as a white dyeable one. DuPont makes white yarn with high recycled-content nylon. Although it may seem like a more sound environmental choice to buy white carpet because it may have higher recycled

content, there are two caveats. First, only a small amount of high recycled-content carpet is currently being made, because only small amounts of recycled nylon is currently available. Second, most white yarn is sold in that form to carpet manufacturers, who dye it themselves.

Peeling Back the Layers

Carpet consists of various layers of materials, and each manufacturer labels these layers a bit differently, depending on its product design and production processes. Fundamentally, carpet is made up of backing; several stabilizing and adhesive layers; and the carpet yarn itself. An adhesive and a peel-off plastic liner can be adhered to the underside of the backing layer to allow for easy installation.

The backing comprises the bulk of the non-yarn portion of the carpet. In some cases, it is an extruded sheet made up of various polymers. In other cases it is a mixture of fillers, resins, and other compounds that give that layer flexibility as well as

strength. The backing sometimes contains the primary recycled content of the carpet. If it is made up of a polymer mixture, these compounds were obtained from industrial sources, sometimes from reclaimed carpet. The fillers are primarily calcium carbonate, some of which is obtained from the backing of used carpet. Some backing contains vinyl from the manufacture of automotive interior parts. Some contains postindustrial recycled polyvinyl chloride (PVC), although it is rare that the PVC portion of carpet backing is recycled. Organic fillers, including carbonates and resins, are used in some carpets. These are usually recycled from the backing in other carpets—which may come from postconsumer sources or from postindustrial sources.

Several secondary layers are built upon the backing. These include adhesives, coats of vinyl sealant, stabilizing reinforcement (fiberglass), and a primary layer into which the actual carpet yarn is tufted. None of these ingredients,

including fiberglass, typically contain recycled content.

Recycling Yarns

Some manufacturers certify their carpet products made with recycled content using independent auditors, and some make marketing claims that have not been proven or confirmed. Carpets may carry labels or tags with claims as to the make-up of a carpet, but consumers should always ask the retailer whether or not the product contains recycled content.

Not surprisingly, many carpet products do not contain any recycled material in the yarn. Those that do, however, tend to have a substantially higher percentage of recycled content percentages in the overall carpet product than those with recycled content only in the non-yarn layers. Recycled content in carpets can range from 20% up to approximately 50%. Carpet with more than 35% recycled content uses

recycled-content yarn. The recycled content of the yarn alone from well-known manufacturers has been verified at 5%–70%, with the higher numbers corresponding to the use of nylon 6.

Reclaiming Carpet

So far, the limiting factor—other than polymer technology—in creating carpet with recycled content has been the availability of returned carpet. Reclamation programs have been growing, however, and key manufacturers have established centers that tend to be located in the southeast in general and Georgia in particular. More information on these centers can be found at www.carpetrecovery.org/landfill_diversion/index.asp.

There are several other good sources for carpet information on the Web. Although it is an industry Web site (for the Carpet and Rug Institute), good general information on carpet, purchasing, installation, and

maintenance can be found at www.carpetbuyershandbook.com. Carpet America Recovery Effort (CARE) is an initiative begun by the carpet industry and government to reduce landfill disposal of old carpet. Information on this program can be found at www.carpetrecovery.org. Finally, information on reclamation programs taking used (postconsumer) carpet can be found at www.carpetrecovery.org/product/CARE_product_grid.pdf.

The problems and issues that have contributed to the carpet industry's reputation for irresponsibility are changing. While environmentally conscious consumers may not yet be able to choose carpet products based on an array of green factors, the industry is working to develop innovative solutions to production and disposal that have been a long time coming. Although carpet recycling is still early in its progress, it is real, and it is expanding. Expect improvements and innovations from this industry.

